Review Intelligence Assistant: architecture

plain code · LLM · local ranker · store · LLM call · decline

OFFLINE — rebuilt daily or weekly; no LLM

Kaggle CSV
515K hotel reviews

Prepare
clean, split by date, chunk by comment

12 BM25 indexes
one per city and sentiment

Train parquet
360K reviews, for statistics

train reviews

Each run writes a new versioned copy; it must pass the golden set, then replicas switch to it one at a time (rollback = old copy).

ONLINE — POST /ask: stateless, scale out with replicas

index · all rows

FastAPI
POST /ask; checks the input (422)

valid

1 · Plan [LLM] [code]
LLM: city, topic, scope. Code: exact values + operation

unknown city or off-topic

filters

2 · Search [ranker]
BM25 in matching indexes, reranked by cross-encoder

reviews

3 · Statistics [code]
pandas over all matching reviews: number or breakdown

stats

4 · Gate [code]
too few or poor matches: decline (no LLM call)

too few or irrelevant matches

evidence

5 · Answer [LLM]
LLM writes the answer
code: IDs, numbers
LLM: claim check

checked

Response
answer, citations, evidence, stats, decline reason

call 1: plan (JSON)

call 2: answer

call 3: claim check

LLM provider (OpenAI-compatible API): one model, three calls per question
One model and one setting for all calls (gpt-4o-mini, Gemini or a local model). Each call is separate and sees only its own prompt.
Timeout and retries on every call. If the model is down, the service still returns evidence and statistics.

Declines return a reason and message and never call the LLM for an answer. Too few matches still returns the few found.

Relevance is judged right after the rerank. If an answer check fails, the evidence is still returned with answer: null.

every stage emits timings, counters, decline reasons

OPERATE — observability and evaluation

Logs
one JSON line per request, no question text

Metrics at /metrics
latency per stage, declines, LLM failures

Alerts
errors, latency, failed checks, spend

Evaluation
40-question golden set + LLM judge